

YONAGUNIJIMA, Japan

WMO: 479120

Lat: 24.4667N

Lon: 123.0106E

Elev: 36

StdP: 100.89

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.9	13.9	5.1	5.5	14.2	6.4	6.0	15.0	16.4	15.3	15.4	15.7	10.8	10	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.0	32.0	27.2	31.5	27.0	31.0	26.8	28.0	30.6	27.6	30.3	27.3	30.0	5.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.3	23.2	29.7	26.9	22.7	29.5	26.6	22.2	29.3	90.3	30.8	88.5	30.3	87.0	30.0	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.5	14.0	12.8	DB	11.4	32.9	1.5	0.7	10.4	33.4	9.5	33.9	8.7	34.3	7.6	34.8	(4)
			WB	7.9	28.4	1.3	0.6	7.0	28.9	6.2	29.3	5.4	29.6	4.5	30.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	24.1	18.6	19.1	20.5	23.2	25.6	28.0	29.1	28.8	27.6	25.7	23.3	20.2	(6)
		DBStd	4.19	2.49	2.80	2.65	2.33	1.85	1.53	0.94	0.97	1.30	1.51	1.91	2.46	(7)
		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
		HDD18.3	76	28	23	11	1	0	0	0	0	0	0	1	12	(9)
		CDD10.0	5164	267	255	326	395	482	538	591	582	528	487	398	315	(10)
		CDD18.3	2198	36	45	79	146	224	288	332	324	278	228	149	68	(11)
		CDH23.3	19238	16	44	161	658	1719	3230	4145	3932	2990	1716	548	78	(12)
(14)	Wind	CDH26.7	5656	0	0	2	45	274	1046	1698	1503	838	221	28	1	(13)
		WSAvg	6.6	8.0	7.6	6.9	6.2	5.3	5.5	5.3	6.1	7.6	7.6	8.2		(14)
		PrecAvg	2321	198	157	157	153	191	167	133	215	280	248	213	194	(15)
		PrecMax	4360	435	319	393	332	592	387	429	531	1109	1106	535	513	(16)
		PrecMin	1173	8	62	45	63	26	33	10	18	29	35	46	34	(17)
		PrecStd	629	95	69	75	68	119	87	99	148	223	222	109	123	(18)
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	24.8	25.8	26.8	28.9	30.7	31.8	32.7	32.6	32.1	30.5	28.6	26.2		(19)
		0.4% MCWB	21.2	22.5	23.5	25.5	26.9	27.6	27.3	27.2	27.2	26.3	24.9	22.6		(20)
		2% DB	23.7	24.6	25.8	27.9	29.6	31.2	32.1	32.0	31.0	29.4	27.5	25.1		(21)
		2% MCWB	20.3	21.4	22.7	24.6	26.2	27.2	27.2	27.1	26.6	25.6	24.0	21.7		(22)
		5% DB	22.8	23.7	25.0	27.1	28.9	30.7	31.6	31.3	30.4	28.6	26.7	24.2		(23)
		5% MCWB	19.6	20.9	22.1	24.2	25.7	27.0	27.0	26.9	26.3	25.0	23.3	20.8		(24)
		10% DB	22.0	23.0	24.2	26.3	28.2	30.1	31.1	30.7	29.8	27.9	25.9	23.4		(25)
		10% MCWB	19.0	20.2	21.4	23.7	25.2	26.8	26.8	26.7	25.9	24.3	22.8	20.2		(26)
(28)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	22.4	23.2	24.3	26.1	27.3	28.2	28.6	28.6	27.8	27.1	25.5	23.7		(27)
		0.4% MCDB	23.8	25.0	26.0	28.0	29.7	30.7	31.3	30.8	30.9	29.5	27.7	25.3		(28)
		2% WB	21.2	22.3	23.4	25.2	26.7	27.7	28.0	28.0	27.2	26.3	24.8	22.6		(29)
		2% MCDB	22.9	24.0	25.3	27.2	28.9	30.2	30.6	30.5	30.1	28.6	26.8	24.3		(30)
		5% WB	20.4	21.5	22.6	24.7	26.2	27.4	27.6	27.5	26.9	25.7	24.1	21.7		(31)
		5% MCDB	22.2	23.3	24.6	26.6	28.2	29.8	30.4	30.2	29.5	28.0	26.0	23.5		(32)
		10% WB	19.6	20.6	21.8	24.1	25.7	27.1	27.2	27.2	26.5	25.1	23.4	20.8		(33)
		10% MCDB	21.6	22.5	23.9	26.0	27.7	29.5	30.1	29.8	29.1	27.4	25.4	22.9		(34)
(35)	Mean Daily Temperature Range	MDBR	3.5	3.6	4.0	3.8	3.9	3.6	4.0	4.1	4.0	3.5	3.3	3.3		(35)
		5% DB MCDBR	4.4	4.5	4.7	4.1	4.2	3.7	4.5	4.7	4.5	4.3	4.1	4.2		(36)
		5% DB MCWBR	3.1	3.3	3.4	2.8	2.4	2.0	2.0	2.1	2.3	2.4	2.6	2.9		(37)
		5% WB MCDBR	3.9	4.1	4.5	4.0	4.0	3.6	4.2	4.4	4.3	4.0	3.8	3.8		(38)
		5% WB MCWBR	3.3	3.5	3.8	2.9	2.5	2.1	2.1	2.2	2.4	2.4	2.7	3.1		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.519	0.541	0.632	0.638	0.557	0.509	0.487	0.507	0.508	0.517	0.514	0.524		(40)
		taud	1.975	1.953	1.763	1.804	2.051	2.213	2.284	2.211	2.180	2.092	2.068	1.977		(41)
		Ebn at Noon	724	744	699	707	764	795	813	798	789	757	726	699		(42)
		Edh at Noon	163	178	225	220	171	145	135	145	146	153	148	157		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.04	2.53	3.22	3.88	4.78	5.56	6.40	5.79	4.98	3.79	2.57	2.03		(44)
		RadStd	0.38	0.50	0.50	0.56	0.65	0.50	0.40	0.46	0.49	0.47	0.32	0.33		(45)

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	+0.38	+0.31	+0.35	N/A	N/A	N/A	N/A	(46)
(47)	Regional (12 neighbors)	+0.27	N/A	N/A	+0.37	+0.26	+0.31	N/A	N/A	+97	+86	(47)

Nomenclature: See separate page